

One private clinic in Cairo. One doctor. One secretary. Evening hours.

- The doctor works in a hospital by day and opens the clinic in the evening, so he often arrives late.
- The secretary books by phone (5 of 6 surveyed clinics book mainly by phone) and runs the waiting room.
- Patients without a booking walk in. Everyone is told "come at opening".
- The doctor's lateness is invisible to patients: they wait at the clinic, not at home.

6 clinics asked

Colleague survey, n = 6 (5 Cairo cardiology, 1 Sharqia urology). A small sample, shown as such.

Nowa is built for this clinic: **نوا** tells each patient when to leave home.

Everyone arrives at opening and waits. The phone rings all evening.

COLLEAGUE SURVEY, N = 6

Time a patient waits in the waiting room:

under 30 min	1 clinic
30 to 60 min	3 clinics
1 to 2 hours	1 clinic
over 2 hours	1 clinic

Median about 45 min, mean about 66 min. Both use assumed range midpoints (20, 45, 90, 150 min).

COLLEAGUE SURVEY, N = 6

The secretary, every evening:

Doctor starts late	5 of 6
Calls per day under 10	5 of 6
Calls per day 30 to 50	1 of 6
Top call topic: booking	5 of 6

Doctor starts late: 15, 30 or 60 min in 5 clinics, on time in 1. Same questions again and again: booking, prices, times, address.

SIMULATED, QUEUE-SIM, 300 EVENINGS If all 30 booked patients arrive at opening, the simulated median wait is 229 min (assumed inputs).

Patients stay home until it is really their time.

- 1 The patient books in the clinic chat and gets a **day and a queue number**, with a private link and a Telegram link.
- 2 The doctor leaves the hospital and taps "في الطريق" (on my way). Nobody is told to leave before this tap.
- 3 At the right minute the patient gets a Telegram message: "انزل دلوقتي" (leave now), from the doctor's pace and the live drive time.
- 4 The chat answers from the clinic's own page and approved health pages, with the source.

THE BOUNDARY

The AI only talks.
Code decides.

Booking, queue order, times and every message are plain code. The AI never books, cancels, sends or changes a time.

4 / IMPACT, WITH A LABEL ON EVERY NUMBER

Simulated: median wait 229 min → 21 min. Today's clinics report 45 to 66 min.

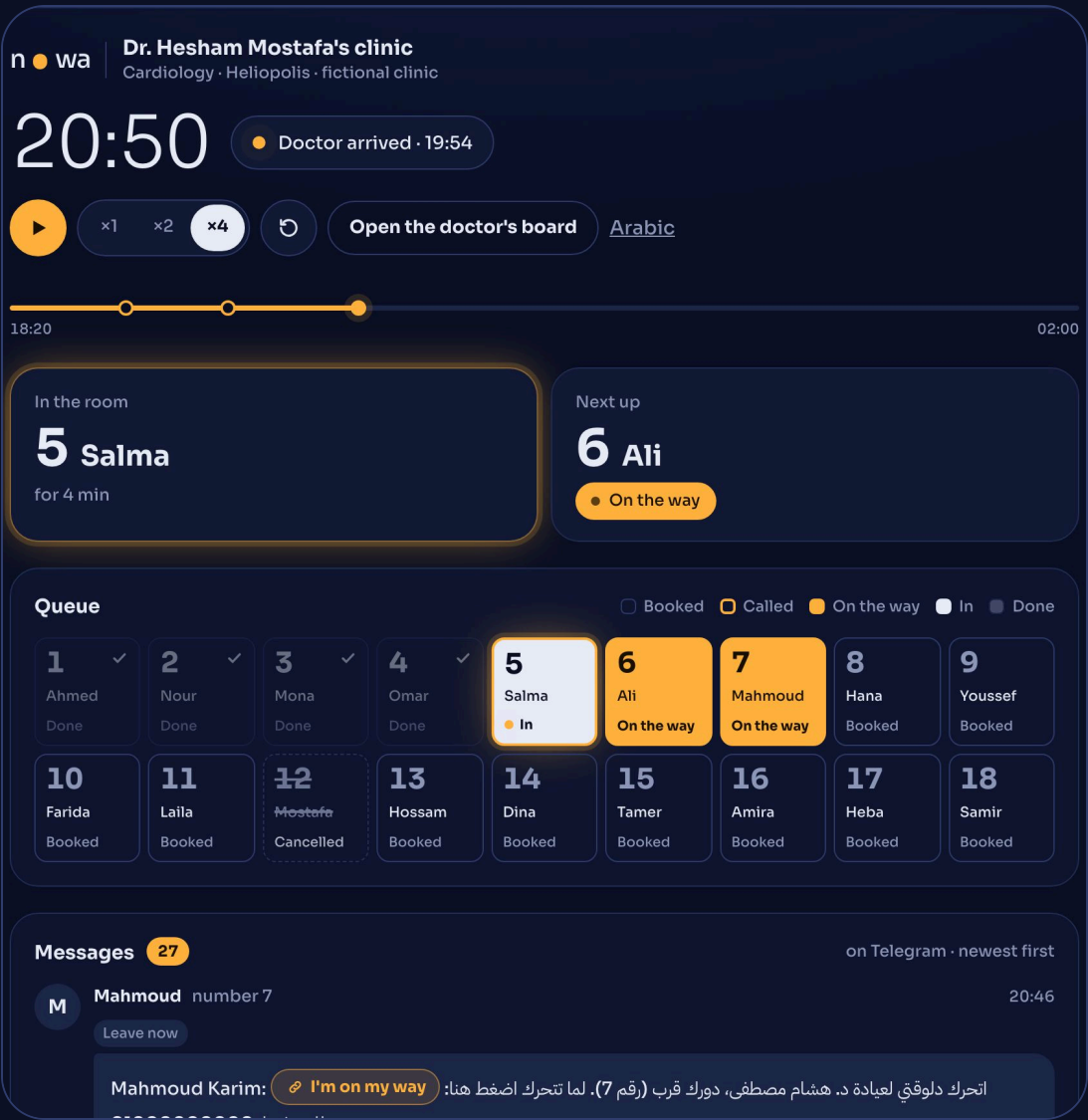
Everyone arrives at opening 229 min median SIMULATED, 300 EVENINGS Colleague survey today: 45 min median, mean ~66 (n = 6)	Patient wait, with Nowa ~21 min median SIMULATED, QUEUE-SIM, 300 RUNS DOCTOR IDLE ~31 MIN/EVENING	
Patient-hours returned, per clinic per month 100–190 h ASSUMED 15 PATIENTS X 17 EVENINGS SIMULATED WAIT	Secretary phone load under 10 calls/day SURVEY, 5 OF 6 CLINICS Nowa answers those questions in the chat; the replaced share is not measured yet	
COST PER PATIENT ~1.7 EGP ESTIMATE	FIXED COST PER MONTH ~1,700 EGP ESTIMATE	PROPOSED PRICE PER CLINIC PER MONTH 1,500 EGP PROPOSED, NOT DECIDED

Sources: research/impact-estimate-2026-09-29.md, costs.md, queue-sim FINDINGS (round 15b), colleague survey 2026-09-28. Cost per patient is the AI chat line only (about 4 messages, a guess); Telegram and Mapbox are free at this size. Revenue recovered is not claimed.

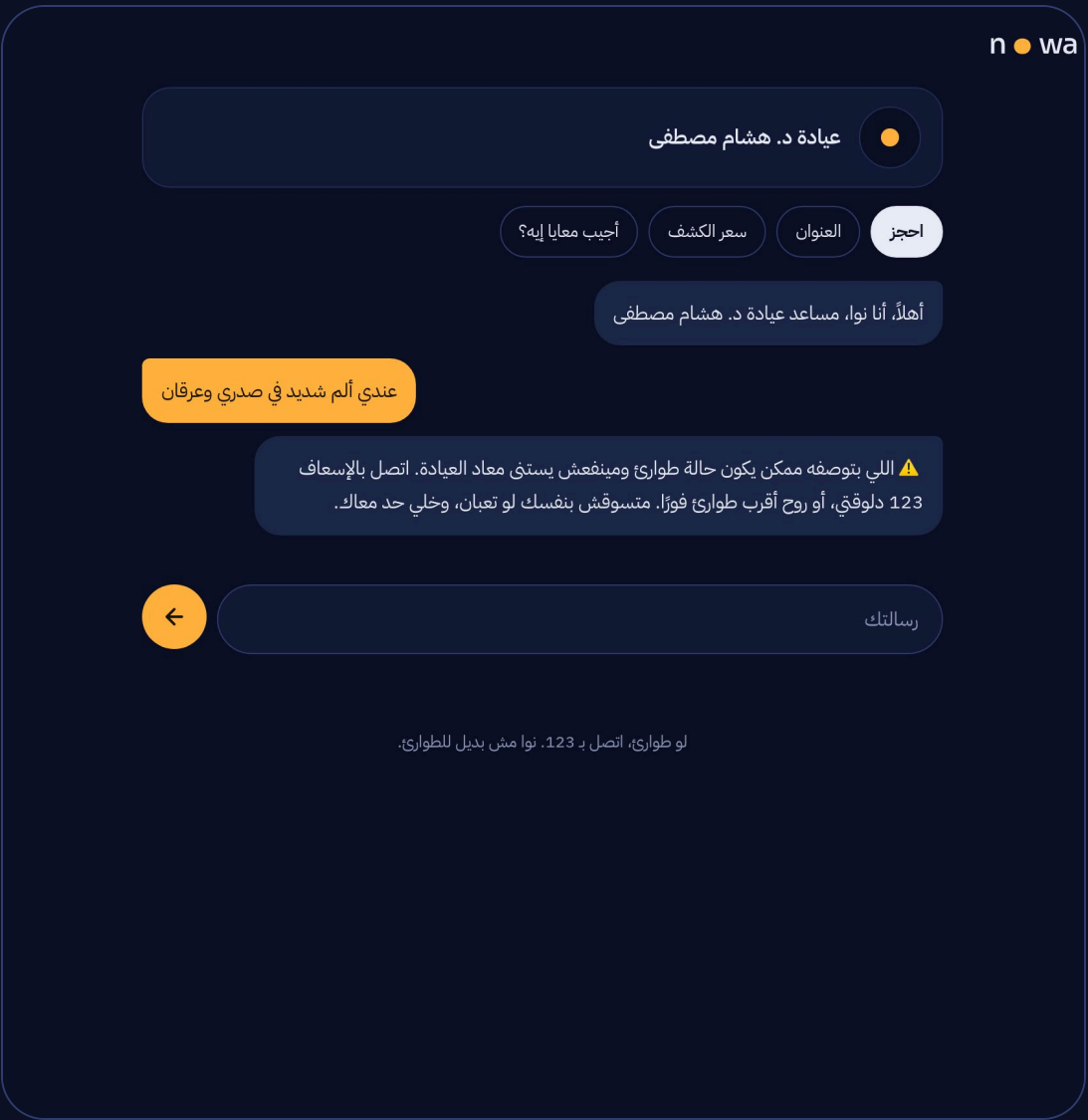
Three things to see, one command to run.



The patient books in the chat and the Telegram message lands.



The doctor taps and the queue moves.



Chest pain: the 123 message appears and booking stops.

Runs in 5 minutes from the repo, no keys: `python -m nowa demo`

Emergencies first. The AI never acts.

Emergency layer first. Every chat is checked by fixed emergency rules before anything else. Chest pain, stroke signs, breathing trouble: a fixed 123 message and booking stops.

The AI never books or sends. It has no tools that act. Patient text cannot act as a command.

Six fixed message templates. Every message is one of six, filled from the database in the patient's language, sent by code.

Egypt Law 151 retention design. Chat text 30 days, event rows 90 days, patient file 3 years after the last visit. Locations are never stored.

Hashed phones. Phones are stored as keyed hashes outside the identity tables. Everything else uses internal IDs.

Telegram only. One channel for patients and doctors. No other messaging vendor.